

Decent Coaching Classes
IX CBSE
Mind Your Math - IV
Answer Key

Section A – Multiple Choice Questions

(5)

1. (B) 1
2. (A) $(a - b)(a^2 + ab + b^2)$
3. (B) bisects
4. (C) 4 [Sample space: HH, HT, TH, TT]
5. (C) 0

Section B – Very Short Answer Questions

(8)

1. $p(y) = 3y^2 - 2y + 7$

$$p(-1) = 3(-1)^2 - 2(-1) + 7 = 3(1) + 2 + 7 = 3 + 2 + 7 = 12$$

2. Chord = 16 cm, so half-chord = 8 cm; distance from centre, $d = 6$ cm.

Since the perpendicular from the centre bisects the chord, radius (r), half-chord and d form a right triangle:

$$r^2 = 8^2 + 6^2 = 64 + 36 = 100$$

$$\therefore r = 10 \text{ cm}$$

3. Experimental probability of an event = (number of times the event occurs) $\div$ (total number of trials)

$$P(\text{two heads}) = 15 \div 60 = 1/4 = 0.25$$

4. Factorise: $9x^2 + 24xy + 16y^2$

$$= (3x)^2 + 2(3x)(4y) + (4y)^2$$

Using $(a + b)^2 = a^2 + 2ab + b^2$, with $a = 3x$, $b = 4y$:

$$= (3x + 4y)^2$$

OR

$$105 \times 95 = (100 + 5)(100 - 5)$$

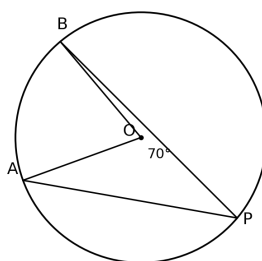
Using $(a + b)(a - b) = a^2 - b^2$, with $a = 100$, $b = 5$:

$$= 100^2 - 5^2 = 10000 - 25 = 9975$$

Section C – Short Answer Questions

(9)

1. Diagram-based question



The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining (major) arc of the circle.

$$\therefore \angle AOB = 2 \times \angle APB$$

$$70^\circ = 2 \times \angle APB$$

$$\therefore \angle APB = 35^\circ$$

2. Total number of marbles = $5 + 7 + 8 = 20$

- (i) $P(\text{red}) = 5/20 = 1/4$
(ii) $P(\text{not blue}) = 1 - P(\text{blue}) = 1 - 7/20 = 13/20$
(iii) $P(\text{red or green}) = (5 + 8)/20 = 13/20$

3. Expand: $(2p - 3q + 4)^2$

Using $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$, with $a = 2p$, $b = -3q$, $c = 4$:

$$= (2p)^2 + (-3q)^2 + (4)^2 + 2(2p)(-3q) + 2(-3q)(4) + 2(2p)(4) \\ = 4p^2 + 9q^2 + 16 - 12pq - 24q + 16p$$

OR

Factorise: $x^3 - 64y^3 = x^3 - (4y)^3$

Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, with $a = x$, $b = 4y$:

$$= (x - 4y)(x^2 + 4xy + 16y^2)$$

Section D – Long Answer Questions

(8)

1. $C(x) = 2x + 50$

(i) $C(10) = 2(10) + 50 = 20 + 50 = 70$. $\therefore$ Total cost = ₹70

(ii) $80 = 2x + 50 \Rightarrow 2x = 30 \Rightarrow x = 15$ GB

(iii) Table of values:

$x = 0 \rightarrow C = 50$; $x = 5 \rightarrow C = 60$; $x = 10 \rightarrow C = 70$; $x = 15 \rightarrow C = 80$

Plotting these points (0,50), (5,60), (10,70), (15,80) and joining them gives a straight line graph.

Reading the graph at $x = 12$: $C = 2(12) + 50 = 74$

$\therefore$ The estimated total cost for 12 extra GB is ₹74.

2. Construction and reasoning

Steps of construction:

(i) Draw a line segment $AB = 6$ cm.

(ii) At A, construct $\angle A = 60^\circ$ using a protractor/compass; at B, construct $\angle B = 75^\circ$.

(iii) Let the two rays meet at C. Join AC and BC to complete $\triangle ABC$.

(iv) Draw the perpendicular bisectors of any two sides (say AB and BC). Let them meet at O — this is the circumcentre.

(v) With O as centre and OA as radius, draw a circle. This circle passes through A, B and C — it is the circumcircle.

Reasoning: $\angle C = 180^\circ - \angle A - \angle B = 180^\circ - 60^\circ - 75^\circ = 45^\circ$.

Since all three angles (60° , 75° , 45°) are less than 90° , $\triangle ABC$ is an acute-angled triangle.

$\therefore$ The circumcentre O lies inside the triangle.

— *End of Answer Key* —